Gap diagnostics (research period only)

Nifty50

outcome	effect bps per 1sd gap	t-stat	n
next tradable day (open t+1 to open t+2)	12.84	3.91	2442
part 1: intraday of day t+1	5.15	1.87	2443
part 2: overnight into t+2	7.68	2.34	2442

Excluding 2020: effect 8.71 bps per 1sd, t=3.34 (n=2192) Winsorised at 1%/99%: effect 10.97 bps per 1sd, t=3.98 Spearman rank correlation gap(t) vs next tradable day: 0.100 (p=0.0000)

By gap quintile:

n_gap	mean next tradable day (bps)	days
Q1 (big down gap)	-6.4	489
Q2	-1.1	488
Q3	0.2	487
Q4	8	489
Q5 (big up gap)	25.6	489

By calendar year:

year	effect bps per 1sd gap	t
2012	10.8	1.16
2013	15.4	1.4
2014	4.2	0.9
2015	13.5	2.33
2016	12.5	2.15
2017	0.1	0.03
2018	-3.1	-0.38
2019	4.3	0.83
2020	31.4	2.56
2021	14.8	2.14

positive in 9 of 10 years

BankNifty

outcome	effect bps per 1sd gap	t-stat	n
next tradable day (open t+1 to open t+2)	18.9	3.05	2442
part 1: intraday of day t+1	10.86	2.58	2443
part 2: overnight into t+2	8.02	2.11	2442

Excluding 2020: effect 9.53 bps per 1sd, t=2.26 (n=2192) Winsorised at 1%/99%: effect 15.29 bps per 1sd, t=3.79 Spearman rank correlation gap(t) vs next tradable day: 0.085 (p=0.0000)

By gap quintile:

b_gap	mean next tradable day (bps)	days
Q1 (big down gap)	-12	489
Q2	2.9	488
Q3	0.8	487

b_gap	mean next tradable day (bps)	days
Q4	10.1	489
Q5 (big up gap)	27.4	489

By calendar year:

year	effect bps per 1sd gap	t
2012	14.6	1.21
2013	15	0.7
2014	2	0.26
2015	22.2	2.8
2016	14.6	1.95
2017	-1.3	-0.24
2018	-5.2	-0.51
2019	1.4	0.17
2020	58.6	2.52
2021	10.7	1.23

positive in 8 of 10 years

Does it hold on both sides? (the rule's own signal: threshold 0.5, window 60)

Nifty50

	mean next tradable day (bps)	t-stat	days
after a big gap up	15.8	4.87	971
after a big gap down	-15.4	-2.2	371
after a small gap	1.9	0.66	1040
all days	4.9	2.19	2382

BankNifty

	mean next tradable day (bps)	t-stat	days
after a big gap up	15.9	3.05	836
after a big gap down	-18	-1.96	405
after a small gap	5.4	1.25	1141
all days	5.1	1.56	2382

Which piece of yesterday predicts tomorrow? (Nifty, joint regression)

n_fwd on gap and intraday together:

	coef	t
const	0	0.978
n_gap	0.203	3.812
n_intra	-0.059	-2.038